

Methodology Workflow: Data → Modeling → Evaluation → Robustness

Phase 1: Data Preparation
PISA 2022 public-use data → Merge student + school questionnaires → Variable audit (missingness < 50%) → Outcome construction (PV_mean, low-performer binary)



Phase 2: Preprocessing
Train/holdout split (80/20, stratified) → Median imputation → One-hot encoding → Standard scaling → Weight normalization



Phase 3a: Baseline Models
Ridge & Elastic Net (regression)
Logistic Regression (classification)

Phase 3b: ML Models
Random Forest · HistGradientBoosting
LightGBM · XGBoost
→ Hyperparameter tuning (Optuna, 50 trials)



Phase 4a: Evaluation
RMSE · MAE · R² (regression)
AUC · F1 · Precision · Recall · Brier (classification)
Calibration: intercept, slope, ECE, decile bins
Threshold sensitivity: Youden's J, max-F1

Phase 4b: Explainability
SHAP summary plots (TreeSHAP, n=5,000)
Permutation importance (n=10,000, 5 repeats)
SHAP interaction values (n=2,000)
Digital-feature importance isolate



Phase 5: Robustness Checks
OECD holdout · Subgroup evaluation (gender, immigrant, ESCS) · Complete-case sensitivity
Plausible-value labels · Country fixed effects · Country-group holdout · Stacking ensemble